SEMESTER 5 – WEEK 2 – LAB EQUIPMENT AND SAFETY

1. Reading Comprehension

Working Safely in the Laboratory

The laboratory is one of the most exciting and dangerous places for a science student. It is where theories are tested, reactions are observed, and discoveries are made. However, without proper safety measures, it can also become a place of serious accidents. Every student must understand that scientific progress depends not only on intelligence but also on discipline and responsibility.

Before entering the laboratory, students should always wear appropriate clothing: a lab coat, safety goggles, and closed shoes. Long hair must be tied back, and jewelry should be avoided. These precautions may seem simple, but they prevent burns, spills, and contamination. One careless moment can ruin hours of work or even harm someone.

In addition to personal safety, cleanliness and organization are essential. Equipment such as beakers, test tubes, pipettes, and microscopes must be used correctly and returned to their places after experiments. Chemical substances should never be mixed without instruction, and labels must always be read carefully. In case of an accident—such as a chemical splash or a small fire—students should immediately inform the supervisor rather than panic or try to hide the problem.

Teamwork is also part of laboratory safety. Each member of a group should have a specific role and remain alert. Communication, cooperation, and mutual respect create a safe working environment. Remember: science aims to improve life, not endanger it. A good scientist is not only curious but also cautious.

A. General Understanding

1. What is the main purpose of the text?
2. Why does the author describe the lab as both exciting and dangerous?

B. True / False / Not Mentioned

Statement	T / F / NM
Wearing safety goggles is optional.	F
Students should report any accident to the supervisor.	T
Jewelry helps protect students from burns.	NM
Teamwork contributes to safety in the lab.	T

C. Multiple Choice

1. What does the author say is *as important as intelligence* in the lab?
   a) Creativity b) Responsibility
   c) Curiosity d) Speed

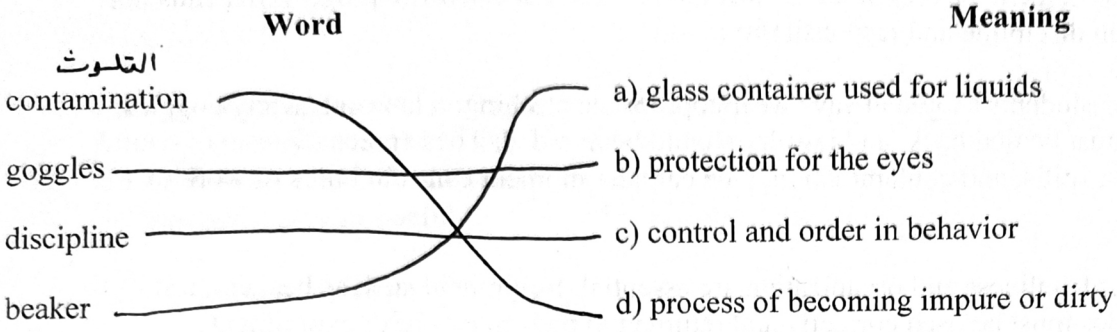
2. What should students do after finishing an experiment?
   a) Leave the equipment on the table
   ✓✗b) Wash all materials and put them back in place✓
   c) Throw away used chemicals
   d) Leave immediately

D. Short Answer

1. Mention two safety rules from the text.
2. What should students do if an accident occurs?

2. Vocabulary Exercises

A. Match the Word to Its Meaning



B. Label the Equipment

Write the correct name under each description:

1. Used to heat liquids over a flame → _____
2. Used to hold small samples → _____
3. Used to measure exact volumes → _____
4. Used to observe tiny organisms → _____

(Answer choices: test tube, microscope, pipette, Bunsen burner)

C. Word Formation

Verb	Noun	Adjective
protect	protection	protective
measure	Measurement	Measurable
observe	observation	observal - observent

3. Grammar – Countable and Uncountable Nouns + Imperatives

A. Explanation

Countable nouns → we can count (one beaker, two beakers).
Uncountable nouns → cannot be counted (water, equipment, information).

Examples:

- *We need three test tubes.* (countable)
- *There is some water in the flask.* (uncountable)
- *Don't touch the hot glass!* (imperative = direct instruction)
- *Always wear your gloves!* (imperative = safety command)

Imperatives are used to give instructions or warnings in labs.
Form: *Base verb + object* (no subject).
Examples:

- *Clean the table after the experiment.*
- *Do not mix unknown chemicals.*

B. Practice 1 – Countable or Uncountable?

Circle the correct form.

1. There is some (equipment / equipments) on the table.
2. We need more (beaker / beakers) for the experiment.
3. She spilled some (chemical / chemicals) on the floor.
4. The professor gave us useful (information / informations).

C. Practice 2 – Imperatives

Complete the safety instructions with the correct imperative form.

1. _____ (wear) your lab coat before starting.
2. _____ (not / eat) or drink in the laboratory.
3. _____ (clean) the table after finishing.
4. _____ (report) any accident immediately.

4. Speaking:

In pairs or small groups, discuss:

- What lab safety rule do you think is the most important? Why?
- Have you ever witnessed or heard of a lab accident? What happened?
- How can teamwork improve safety in the laboratory?

5. Writing:

Write a short report (150–180 words):

"A Day in the Laboratory: How We Keep It Safe and Organized."

Include:

- Description of 3–4 pieces of equipment
- 2–3 safety rules
- At least 2 **imperative sentences**
- At least 2 examples of **countable and uncountable nouns**